

# Linear Regression

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Md. Aminul Islam Shazid

# Outline

- 1 Data as Vectors and Matrices
- 2 Simple Linear Regression
- 3 Multiple Linear Regression
- 4 Prediction Interval
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- 6 Interaction

# Data as Vectors and Matrices

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# Observations, features and labels

| $i$      | features |          |           | label    |
|----------|----------|----------|-----------|----------|
|          | TV       | radio    | newspaper | sales    |
| 1        | 230.1    | 37.8     | 69.2      | 22.1     |
| 2        | 44.5     | 39.3     | 45.1      | 10.4     |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$  | $\vdots$ |
| 200      | 232.1    | 8.6      | 8.7       | 13.4     |

- Each row is one **observation** (a market):  $n = 200$  observations
- The columns TV, radio, newspaper are the **features** (predictors, inputs, independent variables): budgets in thousands of dollars
- Number of columns/features/predictors:  $p = 3$
- The column sales is the **label** (response, output, target, dependent variable): what we want to predict, in thousands of units

# Vectors and matrices

- Observation  $i$  is a **feature vector**  $\mathbf{x}^{(i)} = (x_1^{(i)}, x_2^{(i)}, \dots, x_p^{(i)})^\top \in \mathbb{R}^p$  with label  $y^{(i)} \in \mathbb{R}$ ; for example  $\mathbf{x}^{(1)} = (230.1, 37.8, 69.2)^\top$  and  $y^{(1)} = 22.1$
- Stacking the  $n$  observations gives the **feature matrix**  $\mathbf{X} \in \mathbb{R}^{n \times p}$ , whose row  $i$  is  $(\mathbf{x}^{(i)})^\top$ , and the **label vector**  $\mathbf{y} \in \mathbb{R}^n$

$$\mathbf{X} = \begin{bmatrix} 230.1 & 37.8 & 69.2 \\ 44.5 & 39.3 & 45.1 \\ \vdots & \vdots & \vdots \\ 232.1 & 8.6 & 8.7 \end{bmatrix} = \begin{bmatrix} (\mathbf{x}^{(1)})^\top \\ (\mathbf{x}^{(2)})^\top \\ \vdots \\ (\mathbf{x}^{(n)})^\top \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 22.1 \\ 10.4 \\ \vdots \\ 13.4 \end{bmatrix}$$

- Superscript  $(i)$  = which observation (row), subscript  $j$  = which feature (column):  $x_j^{(i)}$  is entry  $(i, j)$  of  $\mathbf{X}$ ; bold lowercase = column vector, bold uppercase = matrix, plain letter = single number

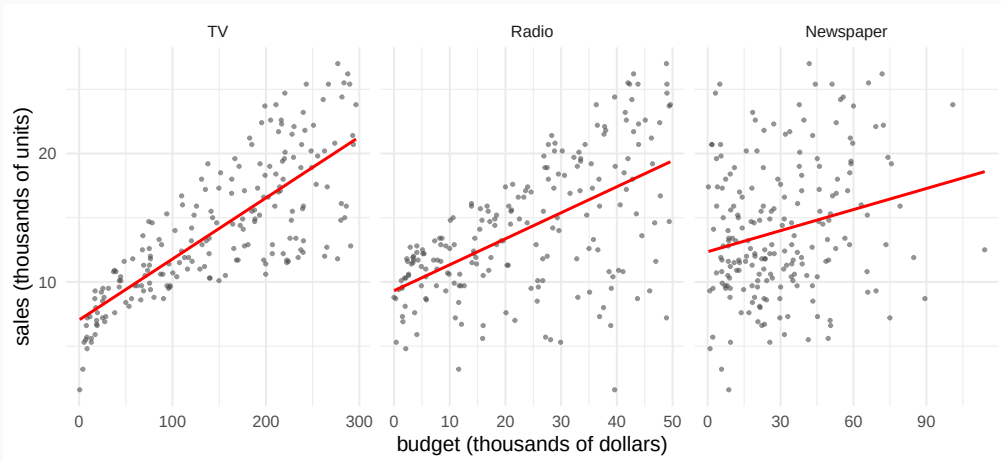
# The regression problem

- Assume the label depends on the features through  $y = f(\mathbf{x}) + \varepsilon$ :  $f$  is a fixed but unknown function and  $\varepsilon$  is a random error with mean zero, independent of  $\mathbf{x}$
- **Supervised learning**: estimate  $f$  from the labelled data  $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$ , which gives an estimate  $\hat{f}$
- **Regression** when the label is quantitative (sales, mpg, balance); classification when it is qualitative (later lectures)
- **Prediction**: compute  $\hat{y} = \hat{f}(\mathbf{x})$  for new observations
- **Inference**: understand how the label changes with each feature (which features matter, how large the effects are, whether the relationship is linear)

# Advertising data: questions regression can answer

- Is there a relationship between the advertising budgets and sales, and how strong is it?
- Which media are associated with sales, and by how much does each contribute?
- How accurately can we predict sales for given budgets?
- Is the relationship linear, and do the media reinforce each other (synergy)?

# Advertising data: questions regression can answer (cont.)



# Simple Linear Regression

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# Simple linear regression model

- One feature  $x$  and a quantitative label  $y$ , related by a straight line plus error:  $y = \beta_0 + \beta_1 x + \varepsilon$
- $\beta_0$  is the **intercept** (the value of  $y$  when  $x = 0$ ),  $\beta_1$  is the **slope** (the change in  $y$  for a one-unit increase in  $x$ ); together they are the **coefficients**
- Estimate the coefficients from the training data (hat = estimated):  $\hat{\beta}_0, \hat{\beta}_1$ ; predict with  $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$
- **Fitted value** of observation  $i$ :  $\hat{y}^{(i)} = \hat{\beta}_0 + \hat{\beta}_1 x^{(i)}$ ; **residual**:  $e^{(i)} = y^{(i)} - \hat{y}^{(i)}$ ; residuals are computed from data, the errors  $\varepsilon^{(i)}$  are never observed
- Example:  $\text{sales} \approx \beta_0 + \beta_1 \times \text{TV}$

# Assumptions of the linear regression model

- For  $i = 1, \dots, n$ :  $y^{(i)} = \beta_0 + \beta_1 x^{(i)} + \varepsilon^{(i)}$ , with the features  $x^{(i)}$  treated as fixed numbers
- **Linearity, zero-mean errors:**  $\mathbb{E}[\varepsilon^{(i)}] = 0$ , so  $\mathbb{E}[y | x] = \beta_0 + \beta_1 x$ ; this line is the *population regression line*
- **Constant variance, uncorrelated errors:**  $\text{Var}(\varepsilon^{(i)}) = \sigma^2$  for every  $i$  and  $\text{Cov}(\varepsilon^{(i)}, \varepsilon^{(k)}) = 0$  for  $i \neq k$
- **Normality**, needed only for exact small-sample tests and intervals:  $\varepsilon^{(i)} \sim N(0, \sigma^2)$
- Computing  $\hat{\beta}_0, \hat{\beta}_1$  needs none of these; unbiasedness and the standard errors use the first three; how to check them with residuals is the topic of the next lecture

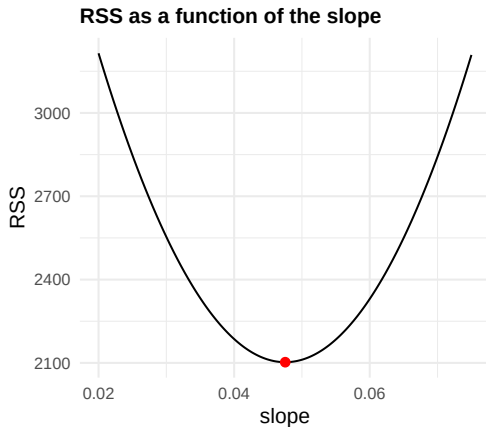
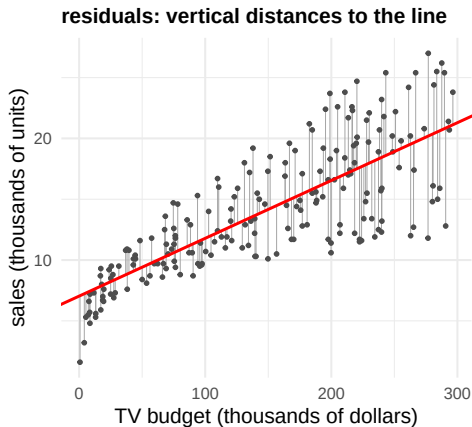
# Least squares criterion

- **Residual sum of squares:**

$$\text{RSS} = \sum_{i=1}^n (e^{(i)})^2 = \sum_{i=1}^n (y^{(i)} - \beta_0 - \beta_1 x^{(i)})^2$$

- **Least squares** chooses  $\hat{\beta}_0, \hat{\beta}_1$  to minimize RSS: the line with the smallest total squared vertical distance to the points
- Squaring stops positive and negative residuals from cancelling and penalizes large ones more; the red dot marks the minimum (RSS = 2102.5 for TV)

# Least squares criterion (cont.)



# OLS for simple regression: first-order conditions

Minimize  $\text{RSS}(\beta_0, \beta_1) = \sum_{i=1}^n (y^{(i)} - \beta_0 - \beta_1 x^{(i)})^2$ . At the minimum both partial derivatives are zero:

$$\frac{\partial \text{RSS}}{\partial \beta_0} = -2 \sum_{i=1}^n (y^{(i)} - \beta_0 - \beta_1 x^{(i)}) = 0$$

$$\frac{\partial \text{RSS}}{\partial \beta_1} = -2 \sum_{i=1}^n x^{(i)} (y^{(i)} - \beta_0 - \beta_1 x^{(i)}) = 0$$

Rearranging gives the **normal equations**, two linear equations in the two unknowns  $\beta_0, \beta_1$ :

$$\begin{aligned} n \beta_0 + \left( \sum_i x^{(i)} \right) \beta_1 &= \sum_i y^{(i)} \\ \left( \sum_i x^{(i)} \right) \beta_0 + \left( \sum_i (x^{(i)})^2 \right) \beta_1 &= \sum_i x^{(i)} y^{(i)} \end{aligned}$$

# OLS for simple regression: solving the normal equations

- Dividing the first normal equation by  $n$  gives  $\bar{y} = \beta_0 + \beta_1 \bar{x}$ , so

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}$$

- Substituting into the second gives

$$\begin{aligned}\sum_i x^{(i)}(y^{(i)} - \bar{y}) &= \hat{\beta}_1 \sum_i x^{(i)}(x^{(i)} - \bar{x}) \\ \hat{\beta}_1 &= \frac{\sum_i x^{(i)}(y^{(i)} - \bar{y})}{\sum_i x^{(i)}(x^{(i)} - \bar{x})} \\ &= \frac{\sum_i (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})}{\sum_i (x^{(i)} - \bar{x})^2} \\ &= \frac{S_{xy}}{S_{xx}}\end{aligned}$$

# Properties of the least squares line

- The line passes through the point of means  $(\bar{x}, \bar{y})$ , by the first normal equation; Advertising:  $7.033 + 0.04754 \times 147.04 = 14.02 = \bar{y}$
- The normal equations say  $\sum_i e^{(i)} = 0$  and  $\sum_i x^{(i)} e^{(i)} = 0$ : the residuals sum to zero and are orthogonal to the feature; consequently  $\sum_i \hat{y}^{(i)} e^{(i)} = 0$  and the fitted values have mean  $\bar{y}$
- The slope is the correlation rescaled:  $\hat{\beta}_1 = r s_y / s_x$ , where  $r$  is the sample correlation and  $s_x, s_y$  are the sample standard deviations; Advertising:  $0.7822 \times 5.217 / 85.85 = 0.0475$
- Since  $|r| \leq 1$ , a one-standard-deviation increase in  $x$  goes with an increase of only  $r$  standard deviations in  $\hat{y}$ : the origin of the name *regression* (towards the mean)

# Interpreting the coefficients

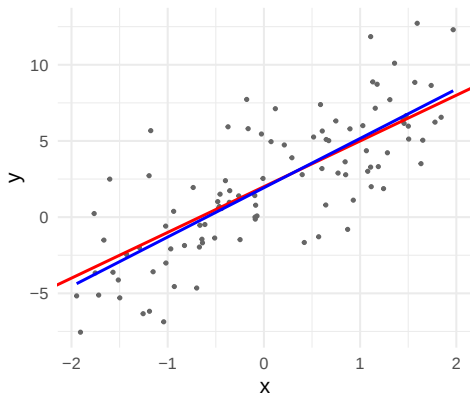
- Fitted line for the Advertising data:  $\widehat{\text{sales}} = 7.03 + 0.0475 \times \text{TV}$  (sales in thousands of units, TV budget in thousands of dollars)
- $\hat{\beta}_1 = 0.0475$ : an additional \$1,000 of TV budget is associated with about  $0.0475 \times 1000 = 47.5$  additional units sold
- $\hat{\beta}_0 = 7.03$ : about 7,030 units are predicted when the TV budget is zero; the smallest TV budget in the data is \$700, so the intercept sits at the edge of the data
- The fit describes association in the observed markets; a causal reading needs more than a fitted line, because budgets were not assigned at random

# How accurate are the estimates?

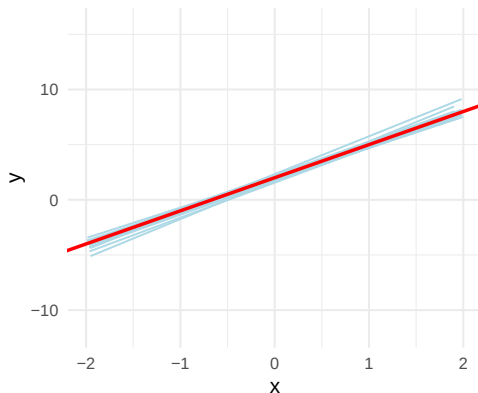
- The population regression line (red) is the true, unknown relationship; the least squares line (blue) is computed from one sample, and every new sample gives a different line
- On average the estimates are correct (unbiased), but a single  $\hat{\beta}_1$  differs from  $\beta_1$ ; the **standard error** measures the typical size of this difference

# How accurate are the estimates? (cont.)

one sample: true line (red), fitted line (blue)



ten samples: ten different fitted lines



# Mean and variance of the slope estimate

- **Unbiased:**  $\mathbb{E}[\hat{\beta}_1] = \beta_1$
- **Variance:**  $\text{Var}(\hat{\beta}_1) = \sigma^2 / S_{xx}$

# Variance of the estimators

$$V(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n (x^{(i)} - \bar{x})^2}, \quad V(\hat{\beta}_0) = \sigma^2 \left[ \frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x^{(i)} - \bar{x})^2} \right]$$

- Here  $\sigma^2 = \text{Var}(\varepsilon)$  is the variance of the errors; standard errors shrink with more data, less noise, and more spread-out  $x^{(i)}$
- $\sigma^2$  is unknown and is estimated from the residuals
- These formulas assume the errors are uncorrelated with constant variance  $\sigma^2$
- The standard errors (SE) of the estimates are found taking the square roots of the variances

# Estimating the error variance

- The errors  $\varepsilon^{(i)}$  are unobserved, so  $\sigma^2$  is estimated from the residuals  $e^{(i)}$
- Recall:  $\text{RSS} = \sum_{i=1}^n (e^{(i)})^2 = \sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2$
- It can be shown that  $\mathbb{E}[\text{RSS}] = (n - 2)\sigma^2$ , therefore,  
 $\hat{\sigma}^2 = \text{RSS}/(n - 2) = \text{MSE}$  is an unbiased estimator of  $\sigma^2$  (here, MSE means mean squared error)
- The residual standard error (RSE) is then given by  
 $\text{RSE} = \sqrt{\text{RSS}/(n - 2)}$
- Two parameters were estimated from the data, so two degrees of freedom are used up; with  $p$  features the divisor becomes  $n - p - 1$
- Advertising, sales on TV:  $\text{RSS} = 2102.53$  and  $n - 2 = 198$  give  
 $\hat{\sigma}^2 = 10.62$  and  $\text{RSE} = 3.26$

# Standard Error of the estimators

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{\text{MSE}}{\sum_{i=1}^n (x^{(i)} - \bar{x})^2}},$$

$$\text{SE}(\hat{\beta}_0) = \sqrt{\text{MSE} \left[ \frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x^{(i)} - \bar{x})^2} \right]}$$

## Confidence intervals and hypothesis tests (recap)

- **95% confidence interval** for  $\beta_1$ : approximately  $\hat{\beta}_1 \pm t_{n-2, \frac{\alpha}{2}} \cdot \text{SE}(\hat{\beta}_1)$
- **Hypothesis test**:  $H_0: \beta_1 = 0$  (no relationship) against  $H_a: \beta_1 \neq 0$ , using  $t = \hat{\beta}_1 / \text{SE}(\hat{\beta}_1)$ , which follows a  $t_{n-2}$  distribution if  $H_0$  is true
- The  **$p$ -value** is the probability of a  $|t|$  at least this large if  $H_0$  were true; a small  $p$ -value (say below 0.05) is evidence against  $H_0$
- Advertising, sales on TV: 95% confidence interval for  $\beta_1$  is [0.0422, 0.0528]

|           | coefficient | std. error | $t$ -statistic | $p$ -value |
|-----------|-------------|------------|----------------|------------|
| intercept | 7.0326      | 0.4578     | 15.36          | < 0.0001   |
| TV        | 0.0475      | 0.0027     | 17.67          | < 0.0001   |

## Assessing the fit: RSE and R-squared

- **Residual standard error:**  $RSE = \sqrt{RSS/(n-2)}$ , the typical size of a residual, in the units of  $y$ ; Advertising:  $RSE = 3.26$ , about 3,260 units (23% of the mean sales)
- $R^2$ : the proportion of the variation in  $y$  explained by the regression

$$R^2 = 1 - \frac{RSS}{TSS}, \quad TSS = \sum_{i=1}^n (y^{(i)} - \bar{y})^2$$

- $0 \leq R^2 \leq 1$ ; Advertising:  $R^2 = 1 - 2102.5/5417.2 = 0.612$ , so TV explains about 61% of the variance in sales
- In simple regression  $R^2 = r^2$ , the squared correlation between  $x$  and  $y$  ( $r = 0.782$ )
- RSE is in the units of  $y$ ,  $R^2$  is unit-free

# Analysis of variance: decomposing the total variation

- Split each deviation from the mean into an explained and an unexplained part:  $y^{(i)} - \bar{y} = (\hat{y}^{(i)} - \bar{y}) + (y^{(i)} - \hat{y}^{(i)})$
- Square and sum over  $i$ ; the cross term  $2 \sum_i (\hat{y}^{(i)} - \bar{y}) e^{(i)}$  is zero because  $\sum_i e^{(i)} = 0$  and  $\sum_i \hat{y}^{(i)} e^{(i)} = 0$  (normal equations), which leaves

$$\underbrace{\sum_{i=1}^n (y^{(i)} - \bar{y})^2}_{\text{TSS}} = \underbrace{\sum_{i=1}^n (\hat{y}^{(i)} - \bar{y})^2}_{\text{ESS}} + \underbrace{\sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2}_{\text{RSS}}$$

- **TSS**: total sum of squares (total variation in  $y$ ); **ESS**: explained (regression) sum of squares; **RSS**: residual sum of squares
- Hence  $R^2 = \text{ESS}/\text{TSS} = 1 - \text{RSS}/\text{TSS}$ ; Advertising, sales on TV:  
 $5417.15 = 3314.62 + 2102.53$

# The ANOVA table for simple regression

| source          | df  | sum of squares | mean square | $F$    | $p$ -value |
|-----------------|-----|----------------|-------------|--------|------------|
| regression (TV) | 1   | 3314.62        | 3314.62     | 312.14 | $< 0.0001$ |
| residual        | 198 | 2102.53        | 10.62       |        |            |
| total           | 199 | 5417.15        |             |        |            |

- Degrees of freedom add up: total  $n - 1 =$  regression 1 + residual  $n - 2$ ; a mean square is a sum of squares divided by its df, so  $MS_{\text{res}} = RSE^2 = 10.62$
- $F = MS_{\text{reg}}/MS_{\text{res}}$  tests  $H_0: \beta_1 = 0$ ; with normal errors  $F \sim F_{1, n-2}$  under  $H_0$

# Multiple Linear Regression

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## Why not several simple regressions?

| sales on  | $\hat{\beta}_1$ | std. error | $t$ -statistic | $p$ -value |
|-----------|-----------------|------------|----------------|------------|
| TV        | 0.0475          | 0.0027     | 17.67          | $< 0.0001$ |
| radio     | 0.2025          | 0.0204     | 9.92           | $< 0.0001$ |
| newspaper | 0.0547          | 0.0166     | 3.30           | 0.0011     |

- Three separate simple regressions give no single prediction from all three budgets, and each one ignores the other media
- A simple-regression slope also absorbs the effect of every correlated feature that was left out
- Newspaper and radio budgets are correlated ( $r = 0.354$ ): markets that spend more on radio tend to spend more on newspaper, so newspaper may look useful only because it stands in for radio
- Remedy: put all features in one model

# Multiple linear regression model

- With  $p$  features:  $y = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p + \varepsilon$ ;  $\beta_j$  is the average change in  $y$  for a one-unit increase in  $x_j$ , *holding all other features fixed*
- For all  $n$  observations at once,  $\mathbf{y} = \tilde{\mathbf{X}}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ :

$$\underbrace{\begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(n)} \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} 1 & x_1^{(1)} & \dots & x_p^{(1)} \\ \vdots & \vdots & & \vdots \\ 1 & x_1^{(n)} & \dots & x_p^{(n)} \end{bmatrix}}_{\tilde{\mathbf{X}}} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}}_{\boldsymbol{\beta}} + \underbrace{\begin{bmatrix} \varepsilon^{(1)} \\ \vdots \\ \varepsilon^{(n)} \end{bmatrix}}_{\boldsymbol{\varepsilon}}$$

- $\tilde{\mathbf{X}} = [\mathbf{1} \ \mathbf{X}] \in \mathbb{R}^{n \times (p+1)}$  is the feature matrix with a column of ones for the intercept (the **design matrix**); fitted values  $\hat{\mathbf{y}} = \tilde{\mathbf{X}}\hat{\boldsymbol{\beta}}$ , residuals  $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$

# OLS for multiple regression: normal equations

Minimize  $\text{RSS}(\beta) = \|\mathbf{y} - \tilde{\mathbf{X}}\beta\|^2 = (\mathbf{y} - \tilde{\mathbf{X}}\beta)^\top (\mathbf{y} - \tilde{\mathbf{X}}\beta)$ . Expanding, and using that the scalar  $\beta^\top \tilde{\mathbf{X}}^\top \mathbf{y}$  equals its transpose  $\mathbf{y}^\top \tilde{\mathbf{X}}\beta$ :

$$\text{RSS}(\beta) = \mathbf{y}^\top \mathbf{y} - 2\beta^\top \tilde{\mathbf{X}}^\top \mathbf{y} + \beta^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \beta$$

With the rules  $\nabla_\beta(\mathbf{a}^\top \beta) = \mathbf{a}$  and  $\nabla_\beta(\beta^\top A\beta) = 2A\beta$  for symmetric  $A$ , the gradient is

$$\nabla_\beta \text{RSS} = -2\tilde{\mathbf{X}}^\top \mathbf{y} + 2\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \beta = \mathbf{0} \quad \implies \quad \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \hat{\beta} = \tilde{\mathbf{X}}^\top \mathbf{y}$$

These are the **normal equations**:  $p + 1$  linear equations in the  $p + 1$  unknowns  $\beta_0, \dots, \beta_p$ .

# OLS for multiple regression: solution

- If  $\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}$  is invertible, the normal equations have the unique solution

$$\hat{\boldsymbol{\beta}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \mathbf{y}$$

# Sampling distribution of the OLS estimator

- Substitute  $\mathbf{y} = \tilde{\mathbf{X}}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  into the solution:  $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \boldsymbol{\varepsilon}$ , so  $\hat{\boldsymbol{\beta}}$  is the truth plus a linear function of the errors
- **Unbiased:**  $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$  gives  $\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$
- **Covariance:** with  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ ,

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top (\sigma^2 \mathbf{I}) \tilde{\mathbf{X}} (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} = \sigma^2 (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$$

- If errors follow the normal distribution, then  $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2 (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1})$ ;  
 $\text{SE}(\hat{\beta}_j) = \hat{\sigma} \sqrt{[(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}]_{jj}}$  with  $\hat{\sigma}^2 = \text{RSS}/(n - p - 1)$ , and  
 $t_j = \hat{\beta}_j / \text{SE}(\hat{\beta}_j) \sim t_{n-p-1}$  under  $H_0: \beta_j = 0$

## Interpreting the coefficients

|           | coefficient | std. error | <i>t</i> - statistic | <i>p</i> - value |
|-----------|-------------|------------|----------------------|------------------|
| intercept | 2.9389      | 0.3119     | 9.42                 | < 0.0001         |
| TV        | 0.0458      | 0.0014     | 32.81                | < 0.0001         |
| radio     | 0.1885      | 0.0086     | 21.89                | < 0.0001         |
| newspaper | -0.0010     | 0.0059     | -0.18                | 0.860            |

- Holding radio and newspaper fixed, an extra \$1,000 on TV is associated with about 45.8 more units sold; an extra \$1,000 on radio with about 188.5 more units
- Newspaper is not significant ( $p = 0.86$ ) once TV and radio are in the model, although it was significant on its own ( $p = 0.0011$ )
- The correlation between newspaper and radio explains it: the simple regression credited newspaper with the effect of radio

## Is at least one feature useful? The F-test

- $H_0: \beta_1 = \beta_2 = \dots = \beta_p = 0$  against  $H_a$ : at least one  $\beta_j \neq 0$

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n - p - 1)} = \frac{\text{ESS}/p}{\text{RSS}/(n - p - 1)}$$

- If  $H_0$  is true,  $F$  is close to 1; a value much larger than 1 is evidence against  $H_0$ , with the  $p$ -value taken from the  $F_{p, n-p-1}$  distribution
- Advertising:  $F = 570.3$  on  $(3, 196)$  degrees of freedom,  $p < 0.0001$ : at least one medium is related to sales
- Why not just look at the individual  $t$ -tests? With many features, about 5% of the  $p$ -values fall below 0.05 by chance even if no feature matters; the  $F$ -test accounts for the number of features
- Deciding *which* features to keep is variable selection, deferred to a later lecture

## The ANOVA table for multiple regression

| source     | df  | sum of squares | mean square | $F$    | $p$ -value |
|------------|-----|----------------|-------------|--------|------------|
| regression | 3   | 4860.32        | 1620.11     | 570.27 | $< 0.0001$ |
| residual   | 196 | 556.83         | 2.84        |        |            |
| total      | 199 | 5417.15        |             |        |            |

- Same layout as in simple regression, now with all three media: regression has  $p$  degrees of freedom, residual  $n - p - 1$ , total  $n - 1$
- $MS_{\text{res}} = \hat{\sigma}^2 = 2.84$ , so  $RSE = 1.69$ ;  
 $ESS = TSS - RSS = 5417.15 - 556.83$
- $R^2 = ESS/TSS = 4860.32/5417.15 = 0.897$ ; the  $F$ -statistic is the same one as on the previous slide

# Model fit in multiple regression

| model                  | $R^2$ | RSE  |
|------------------------|-------|------|
| TV                     | 0.612 | 3.26 |
| radio                  | 0.332 | 4.27 |
| TV + radio             | 0.897 | 1.68 |
| TV + radio + newspaper | 0.897 | 1.69 |

- $RSE = \sqrt{RSS/(n - p - 1)}$  and  $R^2 = 1 - RSS/TSS$ ; here  $R^2$  is also the squared correlation between  $\mathbf{y}$  and  $\hat{\mathbf{y}}$
- Adding radio to TV raises  $R^2$  from 0.612 to 0.897 and cuts RSE from 3.26 to 1.68; adding newspaper changes nothing useful
- $R^2$  never decreases when a feature is added, useful or not, while RSE can increase (the divisor  $n - p - 1$  shrinks); adjusted  $R^2$  corrects this (next lecture)

# Prediction Interval

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# Prediction and its uncertainty

- For a new observation  $\mathbf{x}^* = (x_1^*, \dots, x_p^*)^\top$  that is not in the training data, with  $\tilde{\mathbf{x}}^* = (1, x_1^*, \dots, x_p^*)^\top$ , the prediction is

$$\hat{y}^* = \hat{\beta}_0 + \hat{\beta}_1 x_1^* + \dots + \hat{\beta}_p x_p^* = (\tilde{\mathbf{x}}^*)^\top \hat{\boldsymbol{\beta}}$$

- Uncertainty from estimating the coefficients (reducible):  $\hat{f}$  differs from the true  $f$ ; the **confidence interval** covers the *average* response  $f(\mathbf{x}^*)$  over many observations with these features
- Irreducible error: a single response is  $y^* = f(\mathbf{x}^*) + \varepsilon^*$  even if  $f$  were known; the **prediction interval** covers an *individual* response and is always wider than the confidence interval
- Model bias: a linear model is an approximation of reality; the intervals assume it is correct (checked with residual plots in the next lecture)

# Confidence and prediction intervals: formulas

- The variance of the prediction  $\hat{y}^* = (\tilde{\mathbf{x}}^*)^\top \hat{\boldsymbol{\beta}}$  follows from  $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$ :

$$\text{Var}(\hat{y}^*) = \sigma^2 (\tilde{\mathbf{x}}^*)^\top (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{x}}^*$$

- 95% **confidence interval** for  $f(\mathbf{x}^*)$ :

$$\hat{y}^* \pm t_{0.975, n-p-1} \hat{\sigma} \sqrt{(\tilde{\mathbf{x}}^*)^\top (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{x}}^*}$$

- 95% **prediction interval** for  $y^*$ :

$$\hat{y}^* \pm t_{0.975, n-p-1} \hat{\sigma} \sqrt{1 + (\tilde{\mathbf{x}}^*)^\top (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{x}}^*}; \text{ the extra 1 under the root is the variance of the new error } \varepsilon^*$$

# Confidence and prediction intervals: simple regression

- For a new feature value  $x^*$ , the prediction is  $\hat{y}^* = \hat{\beta}_0 + \hat{\beta}_1 x^*$ , with variance

$$\text{Var}(\hat{y}^*) = \sigma^2 \left[ \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}} \right], \quad S_{xx} = \sum_{i=1}^n (x^{(i)} - \bar{x})^2$$

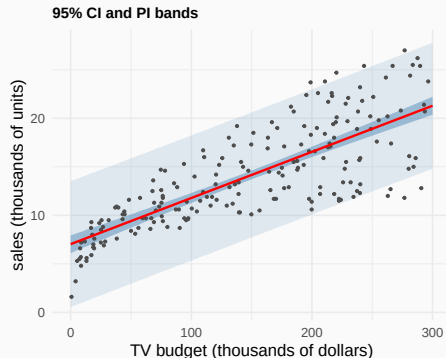
- 95% **confidence interval** for  $f(x^*) = \beta_0 + \beta_1 x^*$ :

$$\hat{y}^* \pm t_{0.975, n-2} \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}}$$

- 95% **prediction interval** for  $y^*$ :  $\hat{y}^* \pm t_{0.975, n-2} \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}}$
- Both are narrowest at  $x^* = \bar{x}$  and widen as  $x^*$  moves away from it; this is the special case  $p = 1$  of the general formulas

# Confidence and prediction intervals for Advertising

- Sales on TV at TV = 100:  
 $\hat{y}^* = 11.79$ , 95% CI [11.27, 12.30],  
95% PI [5.34, 18.23]
- The confidence band is narrowest at the mean TV budget and widens towards the edges; the prediction band stays wide because  $\varepsilon^*$  dominates
- Adding radio = 20 (TV + radio model):  $\hat{y}^* = 11.26$ , CI [10.99, 11.53], PI [7.93, 14.58], narrower because RSE fell from 3.26 to 1.68



# Qualitative Predictors

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## Qualitative features: dummy variables

- Credit data: average credit card balance (dollars) of  $n = 400$  cardholders; do women and men differ?
- Encode gender with a **dummy variable**:  $x^{(i)} = 1$  if person  $i$  is female, 0 if male; the first four cardholders are male, female, male, female:

$$\tilde{\mathbf{X}} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 333 \\ 903 \\ 580 \\ 964 \\ \vdots \end{bmatrix}$$

- Model  $y = \beta_0 + \beta_1 x + \varepsilon$ : the mean balance is  $\beta_0 + \beta_1$  for women and  $\beta_0$  for men, so  $\beta_0$  is the average for men (the **baseline**) and  $\beta_1$  the average difference between women and men

## Qualitative features with two levels: results

- The fitted values are the group means:  $\hat{\beta}_0 = 509.80$  (SE 33.13) for men and  $\hat{\beta}_0 + \hat{\beta}_1 = 509.80 + 19.73 = 529.54$  for women
- $\hat{\beta}_1 = 19.73$  (SE 46.05,  $p = 0.669$ ): women carry about \$20 more, but the difference is not statistically significant
- The  $t$ -test of  $H_0: \beta_1 = 0$  is the two-sample  $t$ -test for equal means with a pooled variance estimate
- Coding male as 1 and female as 0, or as  $\pm 1$ , changes what the coefficients mean but not the fitted values

## Qualitative features with more than two levels

| ethnicity        | $x_1$ (Asian) | $x_2$ (Caucasian) |
|------------------|---------------|-------------------|
| African American | 0             | 0                 |
| Asian            | 1             | 0                 |
| Caucasian        | 0             | 1                 |

- A feature with  $k$  levels needs  $k - 1$  dummy variables; the level without one is the baseline; one 0/1/2 variable would impose an order
- $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ :  $\beta_0$  is the baseline average,  $\beta_1$  and  $\beta_2$  are differences from the baseline
- Credit fit:  $\hat{\beta}_0 = 531.00$ ,  $\hat{\beta}_1 = -18.69$  ( $p = 0.774$ ),  $\hat{\beta}_2 = -12.50$  ( $p = 0.826$ )
- The individual  $p$ -values depend on the baseline; test the whole feature with the  $F$ -test:  $F = 0.04$ ,  $p = 0.957$

# ANOVA for a qualitative feature

| source    | df  | sum of squares | mean square | $F$   | $p$ -value |
|-----------|-----|----------------|-------------|-------|------------|
| ethnicity | 2   | 18,454         | 9,227       | 0.043 | 0.957      |
| residual  | 397 | 84,321,458     | 212,397     |       |            |
| total     | 399 | 84,339,912     |             |       |            |

- Regression on the  $k - 1$  dummy variables reproduces **one-way ANOVA**: the regression sum of squares is the between-group sum of squares and the residual sum of squares is the within-group sum of squares (ANOVA means analysis of variance)
- The  $F$ -test of  $H_0: \beta_1 = \beta_2 = 0$  is the test that all group means are equal; it does not depend on which level is the baseline
- Credit: the group means (531.0, 512.3, 518.5) differ far less than the person-to-person variation within groups, so  $F = 0.043$  and  $R^2 = 18454/84339912 \approx 0.0002$

# Interaction

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# Interactions: relaxing the additive assumption

- The additive model assumes the effect of TV on sales is the same at every radio budget; spending on both together may work better than the sum of the parts (synergy)
- Add an **interaction term**, a new column of  $\tilde{\mathbf{X}}$  with entries  $x_1^{(i)} x_2^{(i)}$ :  
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \varepsilon = \beta_0 + \beta_1 x_1 + (\beta_2 + \beta_3 x_1) x_2 + \varepsilon,$$
so the effect of  $x_2$  depends on  $x_1$

## Interaction: example

- Advertising:  
 $\widehat{\text{sales}} = 6.75 + 0.0191 \text{ TV} + 0.0289 \text{ radio} + 0.0011 \text{ TV} \times \text{radio}$ ; the interaction has  $p < 0.0001$ ,  $R^2$  rises from 0.897 to 0.968 and RSE falls from 1.68 to 0.94
- An extra \$1,000 on radio is associated with about 28.9 more units when  $\text{TV} = 0$  and about 137.5 more units when  $\text{TV} = \$100,000$
- **Hierarchical principle:** if an interaction is in the model, keep the main effects even when their  $p$ -values are large

## Advertising with interaction: model summary

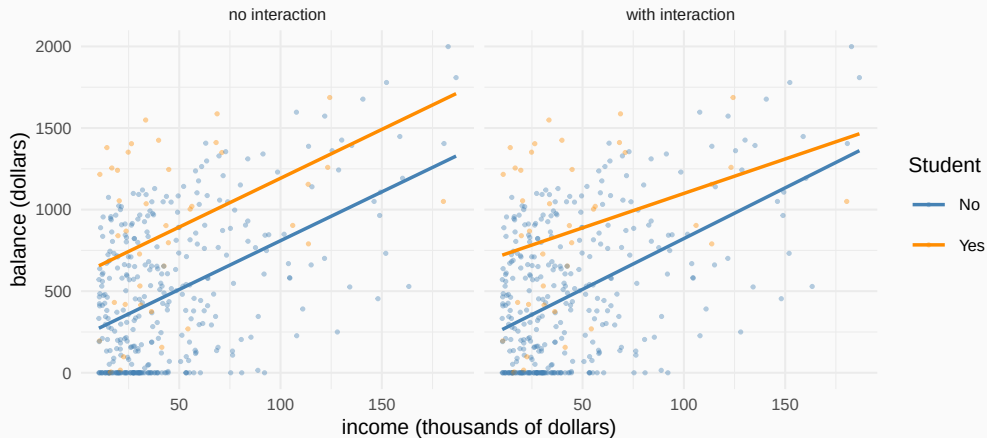
|            | coefficient | std. error | <i>t</i> - statistic | <i>p</i> - value |
|------------|-------------|------------|----------------------|------------------|
| intercept  | 6.7502      | 0.2479     | 27.23                | < 0.0001         |
| TV         | 0.019101    | 0.001504   | 12.70                | < 0.0001         |
| radio      | 0.028860    | 0.008905   | 3.24                 | 0.0014           |
| TV × radio | 0.001086    | 0.000052   | 20.73                | < 0.0001         |

- The interaction is highly significant: the effect of radio on sales grows with the TV budget,  $0.0289 + 0.0011 \text{ TV}$
- $R^2 = 0.968$ ,  $\text{RSE} = 0.944$  on 196 degrees of freedom;  $F = 1963.1$  on  $(3, 196)$ ,  $p < 0.0001$

## Interaction with a qualitative feature

- Credit: balance on income and a student dummy; without the income  $\times$  student interaction the two groups get parallel lines (same slope, different intercept)
- With it, the slope is  $\hat{\beta}_1 = 6.22$  dollars per \$1,000 of income for non-students and  $6.22 - 2.00 = 4.22$  for students
- But the interaction estimate  $-2.00$  has SE 1.73 and  $p = 0.249$ : no strong evidence that the slopes differ

# Interaction with a qualitative feature (cont.)



## Interaction with a qualitative feature: model summary

|                         | coefficient | std. error | t - statistic | p - value |
|-------------------------|-------------|------------|---------------|-----------|
| intercept               | 200.62      | 33.70      | 5.95          | < 0.0001  |
| income                  | 6.218       | 0.592      | 10.50         | < 0.0001  |
| student (yes)           | 476.68      | 104.35     | 4.57          | < 0.0001  |
| income $\times$ student | -1.999      | 1.731      | -1.15         | 0.249     |

- Non-students (baseline):  $\widehat{\text{balance}} = 200.62 + 6.22 \text{ income}$ ; students:  $(200.62 + 476.68) + (6.22 - 2.00) \text{ income} = 677.30 + 4.22 \text{ income}$
- The interaction ( $p = 0.249$ ) is not significant: no strong evidence that the income slopes differ between students and non-students
- $R^2 = 0.280$ ,  $\text{RSE} = 391.6$  on 396 degrees of freedom;  $F = 51.30$  on  $(3, 396)$ ,  $p < 0.0001$

# Notation compared with the ISLP book

This slide index observations with a superscript and features with a subscript. The table maps this to the notation of ISLP (James et al.).

| Quantity                             | This lecture         | ISLP           |
|--------------------------------------|----------------------|----------------|
| Feature $j$ of observation $i$       | $x_j^{(i)}$          | $x_{ij}$       |
| Feature vector of observation $i$    | $\mathbf{x}^{(i)}$   | $\mathbf{x}_i$ |
| Label of observation $i$             | $y^{(i)}$            | $y_i$          |
| Residual of observation $i$          | $e^{(i)}$            | $e_i$          |
| Feature matrix (no intercept column) | $\mathbf{X}$         | $\mathbf{X}$   |
| Design matrix (with column of ones)  | $\tilde{\mathbf{X}}$ | $\mathbf{X}$   |
| New observation                      | $\mathbf{x}^*, y^*$  | $x_0, y_0$     |

**Questions?**

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